University of North Carolina at Charlotte Department of Electrical and Computer Engineering

Homework Report #1

ECGR 4106 - Real-Time Machine Learning

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GitHub: pGovM/introToDL

Problem 1

Objective: Modify the AlexNet architecture, designed for 227x227 ImageNet images and 61+ million paramaters, and simplify it for CIFAR-10's 32x32 images, analyze the result, and explore dropout as a regularization strategy.

Table 1: **Table Detailing all modficiations made to AlexNet**

	Component	Original AlexNet	Modified AlexNet
1.	Conv1 Kernel	11x11	3x3
2.	Conv1 Stride	4	1
3.	Conv2 Kernel	5x5	3x3
4.	Max Pool	3x3, stride 2	2x2, stride 2
5.	Filter counts	96/256/384/384/256	32/64/128/128/64
6.	FC Layers	4096/4096	512/256
7.	Local Response Normalization	Present	Removed

The reason for all the changes listed above are detailed down below:

1. The original convolution kernel would collapse the 32x32 input to a 6x6 spatial map which discards any useful information and structure. The 3x3 kernel with a stride and padding of 1 was chosen to preserve the full 32x32 resolution, information, and structure.
2. As mentioned previously, the stride was changed from 4 to 1 as the input does not need to be as aggressively downsampled as the original 227x227 inputs
3. The second convolution kernel was changed from a kernel size of 5x5 to 3x3. After the first block, the spatial map has been reduced from 32x32 to 16x16, the original kernel would be too large for the reduced map. The 3x3 kernel also simplifies the architecture.
4. The original MaxPool2d() function used a 3x3 overlapping pool which would be too large for CIFAR-10's small spatial maps and the overlapping regions would waste resolution with no advantage. It is replaced with a 2x2 pool with a stride of 2 halves without wasting the resolution as there is no overlap anymore.
5. The most prominent change made was in the filters used in each block. The original architecture used 96 $\rightarrow$ 256 $\rightarrow$ 384 $\rightarrow$ 384 $\rightarrow$ 256 filters respectively. This was due to the complexity and size of the original images which does not apply to CIFAR-10. The modified filter counts start at a lower base and follows the same doubling pattern as follows, 32 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 64. The bottleneck in the last block is to control the number of dimension entering the Fully Connected (FC) layer to reduced the cost of this layer.
6. The FC layer was changed to minimize the flattened dimensions due to the difference in sizes between CIFAR-10 and ImageNet, which also caused the output classes to change from 1,000 to 10 with the number of neurons also changing from 4096 to 256. This also prevents overfitting.
7. Local Response Normalization (LRN) was a normalization technique specific to the original AlexNet and has been obsolete ever since Batch Normalization. Removing it simplifies the AlexNet architecture and does not impede accuracy.

```
Problem          : 1A
Model            : CIFAR10AlexNet (p=0.0)
Epochs           : 40
BatchSize        : 128
Optimizer        : Adam lr=0.001 wd=0.0001
Scheduler        : CosineAnnealingLR T_max=40
Device           : cuda:0
Seed             : 42
```

Figure 1: **Hyper Parameters Used to Train the Model**

```
Modified AlexNet params :      973,322
Original AlexNet params :   61,100,840
Reduction               : 98.4%
```

Figure 2: **Total Countable Parameters between Original and Modified AlexNet**

```
Train/Val Gap Analysis
Variant                 Final TrLoss  Final VlLoss  Gap (Tr-Vl)    VlAcc
─────────────────────────────────────────────────────────────────────────
  Baseline (p=0.0)            0.1964        0.4811      -0.2847   85.84%
  Dropout p=0.3               0.2587        0.4753      -0.2167   85.54%
  Dropout p=0.5               0.2825        0.4860      -0.2034   84.86%

============================================================
PROBLEM 1  Summary
============================================================
  p=0.0 (baseline)      : 85.50% ◄ best
  p=0.3                 : 84.87%
  p=0.5                 : 85.06%
```

Figure 3: **Training and Validation Gap Analysis**

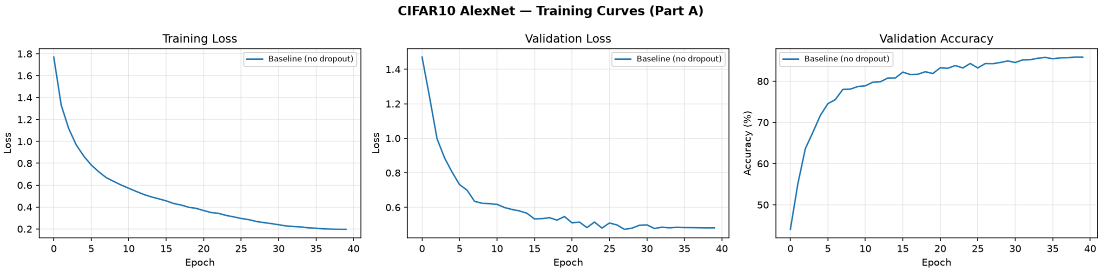


Figure 4: **Training Curves for part A**

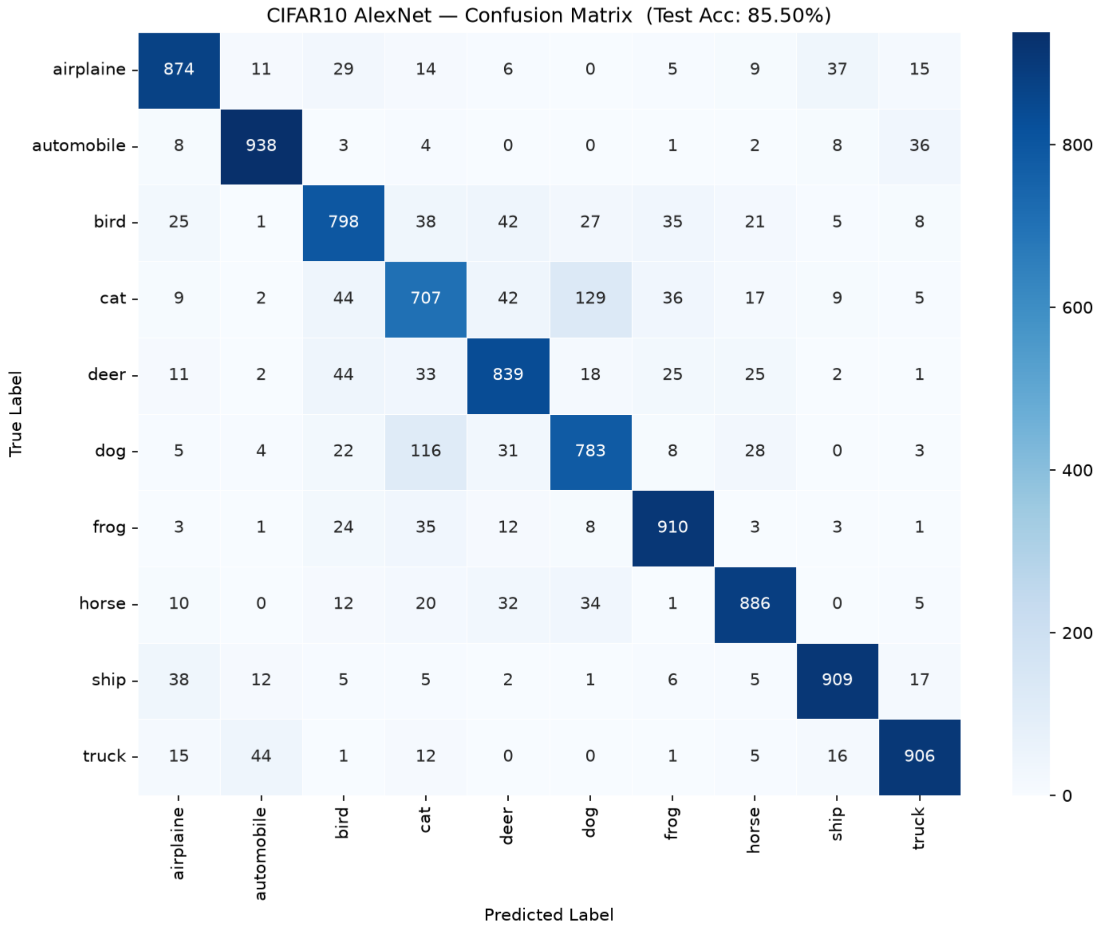


Figure 5: **Confusion Matrix for part A**

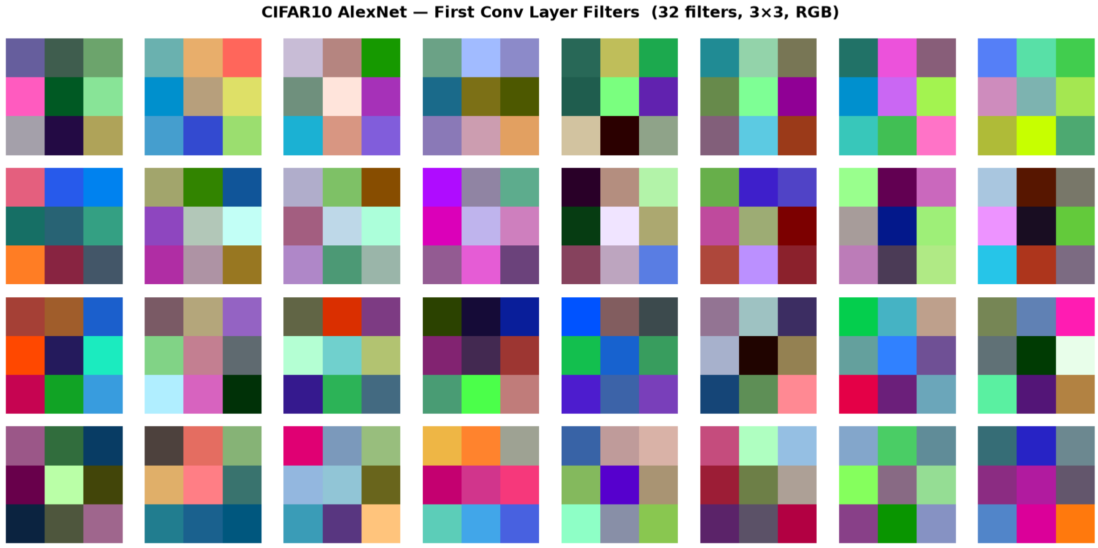
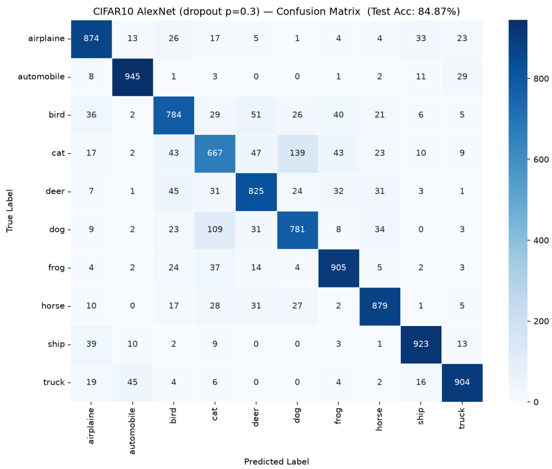
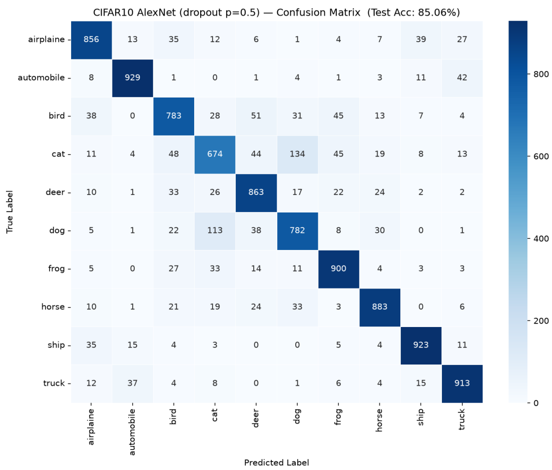


Figure 6: **Filters for part A**



(a) Drouput = 0.3



(b) Drouput = 0.5

Figure 7: Confusion Matrices for Different Dropouts

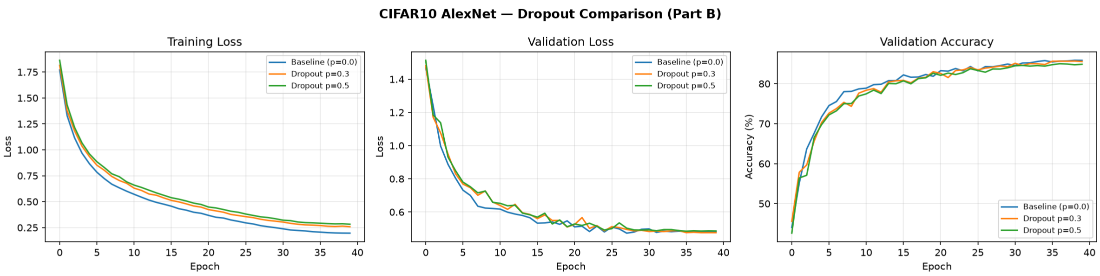


Figure 8: **Comparison between Dropouts**

Problem 2

Objective Pick a suitable VGG configuration (from 11 to 19) whose adapted parameter count is closest to the modified AlexNet, train it under the same conditions, and compare the two architectures.

With the increase in convolution layers stacked between each pool it becomes more difficult to constrain all the additional parameters with the fixed amount of data that CIFAR-10 provides. In addition, the additional layers provide a costly overhead and finer depth that is not required for the CIFAR-10 dataset.

Table 2: **Table Detailing all modficiations made to VGG-11**

	Component	**Original VGG11**	**Modified VGG11**
1.	Filter Scale Factor	N/A	$\div 4$
2.	Filter Cap	512	128
3.	Final Spatial Map	7x7	1x1
4.	Input to FC	512x7x7	128x1x1
5.	FC Progression	$25,088 \rightarrow 4096 \rightarrow 4096 \rightarrow 4096 \rightarrow 1,000$	$128 \rightarrow 1024 \rightarrow 1024 \rightarrow 256 \rightarrow 10$

The reason for all the changes listed above are detailed down below:

1. Due to CIFAR-10 having a lower resolution, the filters can be scaled down to reduce overhead and train the model at a more suitable capacity.
2. At the final two stages, the spatial map is already 2x2 and 1x1 respectively. A higher dimensional flatten would be an unnecessary addition to the overhead cost of training the model.
3. Since there are 8 layers being used on a 32x32 input, the final spatial map, as a direct consequence, will become 1x1 instead of the 7x7 achieved from the ImageNet input.
4. If a bigger input were to be given to the FC, then this layer would dominate and undermine the convolution backbone, due to this a smaller, more reasonable input is given to allow the classifer to combine features without either one being dwarfed.
5. The FC progression was changed for three main reasons. The first being that it had to rebuilt around the 128 convolution layer input, second the number of parameters being entered is not as big as the original ImageNet dataset, and finally, CIFAR-10 only has ten classes while the original had a thousand.

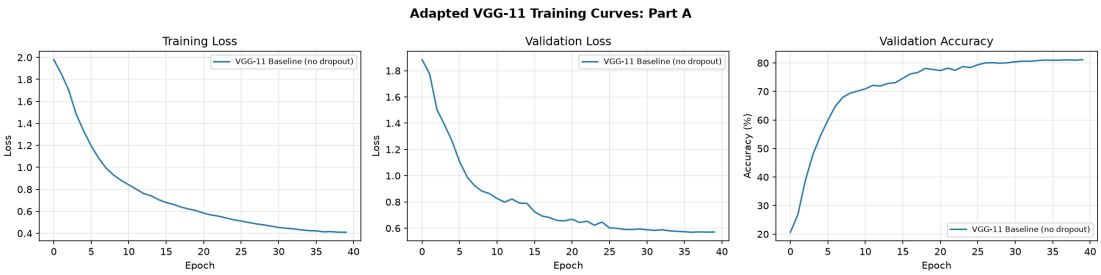


Figure 9: **Training Curves for part A**

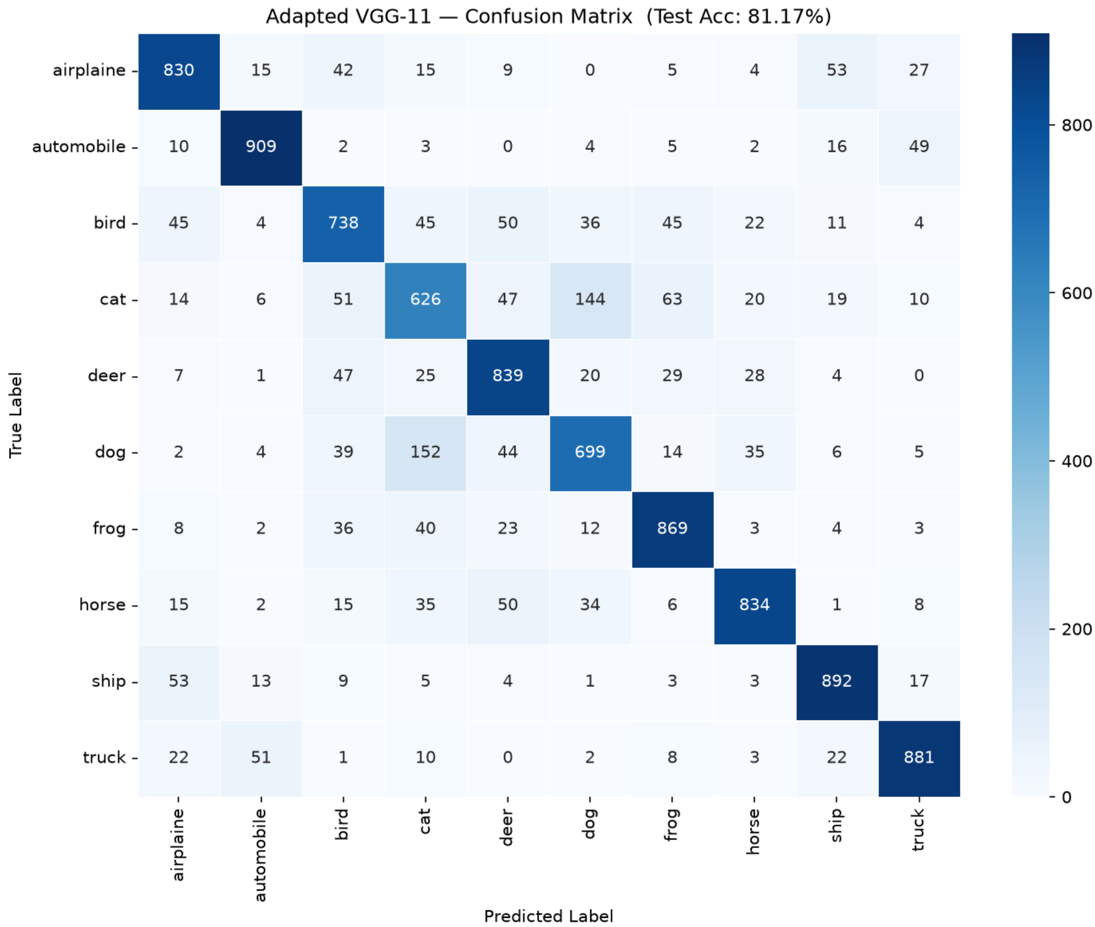
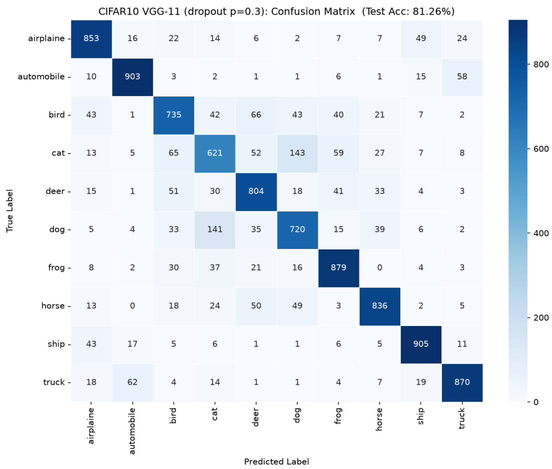


Figure 10: **Confusion Matrix for part A**

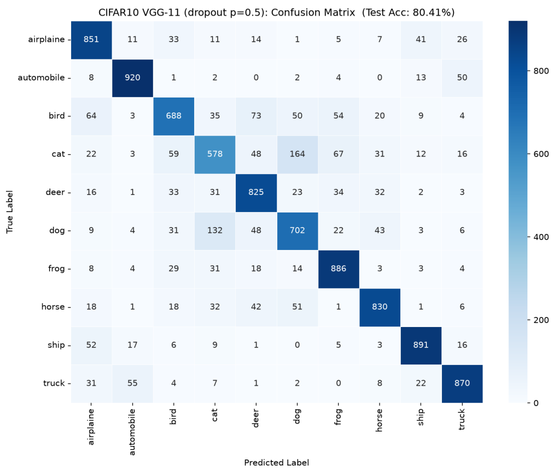
Table 3: **Comparing Modified AlexNet and VGG11**

	Metric	Cifar-10 AlexNet	CIFAR-10 VGG11
1.	Total Parameters	973,322	974,186
2.	Baseline Test Accuracy (%)	85.5	81.17
3.	Best Test Accuracy (%)	85.5	81.26
4.	Avg Epoch Training Time (s)	3.7	3.65
5.	Convergence Speed (No. of epochs)	N/A	8

Since AlexNet only have 5 convolution layers compared to VGG11's 8, it would definitely train faster per epoch along with the difference in the number of sequential layer operations need to be done. Similarly, the increased number of layers allows VGG-11 to generalise better as it is able to build abstract more features and with the former's unform 3x3 kernel design it is able to construct a more consistent feature hierarchy. The structural changes between both architectures that lead to their gap in accuracy and loss are depth, rate of spatial compression, reconstruction and class-score compression, and finally dropout interaction.



(a) Drouput = 0.3



(b) Drouput = 0.5

Figure 11: Confusion Matrices for Different Dropouts

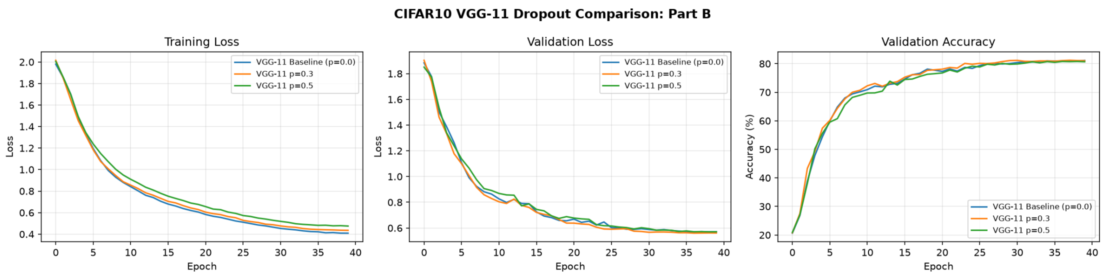


Figure 12: **Comparison between Dropouts**

```
VGG-11 Train/Val Gap Analysis
  Variant                        TrLoss     VlLoss        Gap    VlAcc
  ──────────────────────────────────────────────────────────────────
  VGG-11 Baseline (p=0.0)        0.4094     0.5709    -0.1615   81.14%
  VGG-11 p=0.3                   0.4356     0.5609    -0.1253   81.04%
  VGG-11 p=0.5                   0.4755     0.5706    -0.0951   80.68%

Cross-Model Dropout Summary
  Variant                     Test Acc (%)
  ────────────────────────────────────────
  AlexNet p=0.0                      85.50
  AlexNet p=0.3                      84.87
  AlexNet p=0.5                      85.06
  VGG-11 p=0.0                       81.17
  VGG-11 p=0.3                       81.26
  VGG-11 p=0.5                       80.41
```

Figure 13: **Comparison between Cross-Model Dropouts**

A deeper architecture doesn't respond differently to dropout as the this only affects the FC layer whereas the depth is derived from the convolution layer backbone. However, due to the amount of information retained in VGG-11, the dropout does hurt slightly less compares to AlexNet.

Table 4: **Comparing Best Variant Performance**

Model	Baseline Acc (%)	p=0.3 Acc (%)	p=0.5 Acc (%)
CIFAR-10 AlexNet	85.5 (**BEST**)	84.87	85.06
CIFAR-10 VGG-11	81.17	81.26 (**BEST**)	80.41